TPM2-Related Myopathy (MONDO:0100196): A Comprehensive Disease Characterization

Causal gene: *TPM2* (β-tropomyosin), HGNC:12011, NCBI Gene 7169, Ensembl ENSG00000198467, UniProt P07951, cytoband 9p13.3, OMIM gene 190990.

> **Causal gene stated up front:** This report concerns disease caused by pathogenic variants in **TPM2**, the gene encoding **β-tropomyosin (beta-tropomyosin / slow skeletal-muscle tropomyosin)**, a sarcomeric thin-filament regulatory protein. Serial-number disease aliases (nemaline myopathy 4 / NEM4, congenital myopathy 23 / CMYP23, cap myopathy 2, distal arthrogryposis 1A/2B) all map to TPM2. This is distinct from **TPM3** (α-tropomyosin slow — a different gene causing an overlapping myopathy) and from **TPM1** (cardiac α-tropomyosin — causing cardiomyopathy, not this disease).

Summary

TPM2-related myopathy (MONDO:0100196) is a rare, predominantly autosomal-dominant congenital myopathy of skeletal muscle caused by germline pathogenic variants in ***TPM2***, the gene encoding the sarcomeric thin-filament protein **β-tropomyosin**. This single gene produces an unusually broad clinical–histopathological spectrum: **nemaline myopathy (NEM4 / congenital myopathy 23)**, **cap myopathy**, **core-rod myopathy**, **congenital fiber-type disproportion (CFTD)**, and the contracture-predominant **distal arthrogryposis types 1A and 2B (Sheldon-Hall syndrome)** and **trismus-pseudocamptodactyly (DA7)**. A rare biallelic (recessive) loss-of-function mechanism causes **Escobar-variant multiple pterygium syndrome with congenital myopathy**. The most penetrant single feature across all entities is **muscle weakness**, accompanied by hypotonia, myopathic facies, and (for a subset of alleles) joint contractures.

The dominant pathomechanism is **gain-of-function at the actin–tropomyosin interface**, most prominently manifested as **increased myofibrillar calcium sensitivity ("hypercontractility")**. This molecular phenotype correlates with the clinical contracture phenotype: patients with hypercontractile alleles have far higher rates of limb-joint and jaw contractures. A second, more recently described pathomechanism is **disrupted myogenesis / myofibrillogenesis**, validated across *Drosophila*, mouse myotube, and zebrafish model systems, in which variant severity in developmental assays tracks with clinical severity. Population genetics corroborate a non-haploinsufficiency mechanism: *TPM2* is strongly **missense-constrained** (gnomAD missense Z = 3.81) yet **loss-of-function–tolerant** (LOEUF 0.88, pLI ≈ 0), exactly the signature expected for a dominant missense/dominant-negative disorder rather than one driven by gene dosage.

Clinically, TPM2 disease tends to be **milder and more stable than TPM3-related disease**, with normal or only mildly elevated creatine kinase, slow-to-no progression, and generally normal life expectancy, although **respiratory muscle weakness is the principal threat to survival**, and a cap-myopathy subset can progress to infantile respiratory failure. There is **no approved disease-modifying therapy**; management is supportive and multidisciplinary (respiratory surveillance, orthopedic correction of contractures/scoliosis, physiotherapy, and nutritional support). One notable treatable complication is a **secondary neuromuscular-transmission defect** in some cap-myopathy patients that responds to anticholinesterase therapy. Diagnosis rests on **muscle biopsy histopathology** (nemaline rods, caps, cores, fiber-type disproportion) plus **molecular confirmation by NGS gene panels or exome/genome sequencing**.

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1. Disease Information

TPM2-related myopathy is a **congenital myopathy of the musculoskeletal system** caused by pathogenic variants in the skeletal-muscle β-tropomyosin gene. The MONDO definition describes it as covering *"a wide spectrum of phenotypes … characterized by muscle weakness, amyotrophy, hypotonia, myopathic facies, scoliosis, and sometimes contractures,"* with histology that is *"variable with nemaline and intranuclear bodies, cap-like lesions, core-like lesions, fiber-type disproportion, and dystrophic features."* MONDO notes that **the most penetrant phenotype among all TPM2-related myopathy entities is muscular weakness**, and that contracture-primary distal arthrogryposis is a related but partly distinct entity.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0100196 (TPM2-related myopathy)
Gene	TPM2 · HGNC:12011 · NCBI Gene 7169 · Ensembl ENSG00000198467 · UniProt P07951 · OMIM gene 190990 · cytoband 9p13.3
OMIM (disease aliases)	NEM4 609285 · DA1 108120 · DA2B 601680
Orphanet (entities)	Typical NM ORPHA:171436 · Childhood NM ORPHA:171439 · Cap myopathy ORPHA:171881 · CFTD ORPHA:2020 · DA1 ORPHA:1146 · DA2B/Sheldon-Hall ORPHA:1147
GARD	0026079

Synonyms / alternative names: beta-tropomyosin-related myopathy; TPM2 myopathy; autosomal dominant TPM2-related myopathy; congenital myopathy 23 (CMYP23/CMYO23); nemaline myopathy 4 (NEM4); cap myopathy 2; core-rod myopathy; congenital fiber-type disproportion; distal arthrogryposis 1A and 2B; trismus-pseudocamptodactyly (DA7); Escobar-variant multiple pterygium syndrome (recessive).

Information source: Aggregated disease-level resources (OMIM, Orphanet, MONDO, HPO) plus primary-literature cohorts and case reports — not EHR/individual-patient registries.

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2. Etiology

Primary cause — genetic. TPM2-related myopathy is a **monogenic Mendelian disorder** caused by pathogenic germline variants in *TPM2*. The overwhelming majority are **heterozygous missense substitutions and small in-frame indels** acting dominantly (e.g., ΔK7/p.K7del, ΔK49, R91G, ΔE139, K168E, E181K). A review of **53 families** identified **30 distinct pathogenic variants** with autosomal-dominant inheritance (PMID: 24692096). A rare **recessive** route exists through biallelic loss-of-function (e.g., homozygous splice variant c.564-2A>C) (PMID: 33558124).

Genetic risk factors: The causal variant itself is the risk factor; there are no established susceptibility loci or polygenic modifiers. Variant position (within actin-binding structural motifs vs elsewhere) influences the biochemical and clinical phenotype, though one DA study found "no relation between the types or locations of TPM2 mutations and the clinical characteristics" (PMID: 30285720), tempering simple position-based rules.

Environmental risk factors: None established. This is a constitutional genetic disorder; onset is typically congenital/neonatal and not triggered by environmental exposures. **Consanguinity** is relevant only to the rare recessive form (PMID: 33558124).

Protective factors: No genetic or environmental protective factors have been identified. Notably, gnomAD shows **loss-of-function tolerance** — heterozygous LoF is not disease-causing, which is mechanistically informative (haploinsufficiency is non-pathogenic) rather than "protective."

Gene–environment interactions: None documented. This is essentially a purely genetic disorder.

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3. Phenotypes

TPM2 disease is **skeletal-muscle-restricted**; **cardiac and CNS involvement is NOT characteristic**. Cardinal phenotypes with suggested HPO terms, severity, and onset:

Phenotype	HPO term	Type	Onset	Frequency/severity
Generalized hypotonia	HP:0001290 (neonatal HP:0001319)	Clinical sign	Congenital/ neonatal	Very common; cardinal feature
Muscle weakness (proximal)	HP:0003701; Myopathy HP:0003198	Symptom/ sign	Congenital–childhood	Most penetrant feature
Facial weakness	HP:0030319; Myopathic facies HP:0002058	Clinical sign	Early	Common
Ptosis	HP:0000508	Clinical sign	Early	Frequent
High/arched palate	HP:0000218	Physical	Congenital	Frequent
Feeding difficulties	HP:0011968	Symptom	Neonatal	Common, variable
Respiratory insufficiency (muscle weakness)	HP:0002747	Sign	Variable	Main prognostic threat
Scoliosis	HP:0002650	Physical	Childhood	Common
Joint contractures	HP:0034392	Physical	Congenital	Allele-dependent; hypercontractile alleles
Arthrogryposis multiplex congenita	HP:0002804	Physical	Congenital	DA subtypes
Talipes equinovarus	HP:0001762	Physical	Congenital	DA subtypes
Camptodactyly	HP:0012385	Physical	Congenital	DA subtypes
Ulnar deviation of hand	HP:0009465	Physical	Congenital	DA subtypes
Nemaline bodies (biopsy)	HP:0003798	Lab/ histology	—	Defining in NM form
Central core regions	HP:0030230		—	Core-rod form

Phenotype	HPO term	Type	Onset	Frequency/severity
		Lab/ histology		
Type 1 fiber predominance	HP:0003803	Lab/ histology	—	CFTD form
Myopathic EMG	HP:0003458	Lab	—	Common
Elevated creatine kinase	HP:0008180	Lab	—	Normal to mildly elevated

Severity and progression: Generally **mild and relatively stable**; TPM2 patients *"tended to present with milder symptoms than those with TPM3 mutations"* (PMID: 24692096). Distal-arthrogryposis contractures are typically **non-progressive and congenital** (PMID: 23401156).

Quality-of-life impact: Weakness, contractures, scoliosis, and respiratory/bulbar involvement impair mobility, feeding, and respiratory function; disease-specific QoL instruments (EQ-5D/SF-36/PROMIS) have not been applied to TPM2-only cohorts (knowledge gap).

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4. Genetic / Molecular Information

Causal gene: *TPM2* (β-tropomyosin), 9p13.3, OMIM 190990. Encodes a two-chain coiled-coil thin-filament regulatory protein that, with the troponin complex, controls Ca^{2+}-dependent actin–myosin interaction.

Pathogenic variants: - **Classification:** ClinVar (~480 records): ~148 pathogenic, ~46 likely pathogenic, ~285 VUS, plus benign/likely-benign. - **Variant type/class:** Overwhelmingly **missense and small in-frame indels** (e.g., ΔK7/c.20_22del, ΔK49, R91G, ΔE139, K168E, E181K, Q103R). Rare splice/intronic LoF (c.564-2A>C) in the recessive form. - **Allele frequency:** Pathogenic variants are private/rare; absent or extremely rare in gnomAD. - **Origin:** Germline; both inherited and de novo (recurrent de novo missense documented in MPS-related disorder, PMID: 32092148). No somatic role. - **Functional consequences: Gain-of-function / dominant-negative** — increased myofibrillar Ca^{2+} sensitivity and/or disrupted thin-filament polymerization and myogenesis. Recessive disease = **loss of function** (exon-6b isoform).

Population constraint (gnomAD v2.1.1): missense Z = **3.81** (strong missense depletion); **pLI ≈ 5.5×10^{-6}**, observed/expected LoF = 0.63 (26 obs / 41 exp), **LOEUF = 0.88** — the gene is **missense-constrained but NOT loss-of-function–intolerant**, the hallmark of a dominant missense/dominant-negative disorder, not haploinsufficiency.

Modifier genes / epigenetics / chromosomal abnormalities: None established for TPM2 disease. No DNA-methylation/histone signatures identified; no recurrent structural/cytogenetic lesions.

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5. Environmental Information

Environmental factors: None implicated — this is a constitutional genetic disorder. **Lifestyle factors:** None causative; physiotherapy and activity modification are supportive, not etiologic. **Infectious agents:** Not applicable. (Note: sporadic late-onset nemaline myopathy and HIV-associated nemaline myopathy are separate acquired entities not caused by TPM2.)

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

Dominant (common) branch — gain-of-function:

1. A **heterozygous germline missense or small in-frame indel in *TPM2*** (e.g., ΔK7, R91G, E139K) **alters the β-tropomyosin coiled-coil**, especially at residues in repeating actin-binding structural motifs or the N-terminal head-to-tail polymerization region. *(demonstrated: genetics + biochemistry)*
2. This **leads to** altered positioning/dynamics of tropomyosin on the actin thin filament at the **actin–tropomyosin interface**. *(demonstrated in vitro)*
3. Which **results in** either **(a) increased myofibrillar Ca^{2+} sensitivity (hypercontractility)** or **(b) disrupted head-to-tail polymerization and thin-filament assembly**. *(demonstrated: in-vitro motility; zebrafish sarcomere assays)*
4. Branch (a) **leads to** excess contractile tone in developing muscle → **fixed joint and jaw contractures** (distal arthrogryposis, trismus-pseudocamptodactyly). *(inferred from genotype–phenotype correlation)*

5. Branch (b) **leads to disrupted myogenesis/myofibrillogenesis** → abnormal fiber formation and structural lesions (nemaline rods from Z-disk/α-actinin material; cap-like subsarcolemmal accumulations; cores; fiber-type disproportion). *(demonstrated in fly/mouse/fish; severity correlates with clinical severity)*
6. Both branches **result in muscle weakness, hypotonia, myopathic facies, feeding/respiratory difficulty, scoliosis**. *(clinical)*
7. In some cap-myopathy patients, thin-filament disorganization is **associated with** a **secondary neuromuscular-transmission defect** → fatigable, anticholinesterase-responsive weakness. *(demonstrated in a case)*

Recessive (rare) branch — loss-of-function:

1′. **Biallelic LoF** (homozygous splice variant c.564-2A>C abolishing exon-6b acceptor) **leads to** loss of the exon-6b skeletal β-tropomyosin isoform → **Escobar-variant multiple pterygium syndrome + congenital myopathy**; heterozygotes unaffected. *(demonstrated)*

Upstream vs downstream

Level	Mechanism	Evidence class
Upstream (initiating)	TPM2 missense/in-frame indel altering coiled-coil	Human genetics; gnomAD constraint
Proximal molecular	Altered actin–tropomyosin interface; ↑ Ca^{2+} sensitivity; impaired polymerization	In vitro motility; computational
Cellular/developmental	Disrupted myofibrillogenesis; rods/caps/cores	Fly/mouse/fish models; biopsy
Tissue/physiology	Hypercontractility, weakness, fatigue	Clinical; electrophysiology
Downstream (clinical)	Contractures, hypotonia, respiratory/bulbar weakness, scoliosis	Clinical cohorts

Molecular pathways / processes (GO): muscle contraction (**GO:0006936**), actin filament organization (**GO:0007015**), regulation of ATP-dependent activity (**GO:0043462**), myogenesis/muscle development. **Molecular function:** actin filament binding (**GO:0051015**), structural constituent of muscle (**GO:0008307**), protein homo-/heterodimerization (**GO:0042803**/**GO:0046982**).

Cell types / subcellular (CL / GO-CC): slow type-I skeletal muscle fiber (**CL:0002211**), skeletal muscle fiber (**CL:0000188**); sarcomere (**GO:0030017**), muscle thin-filament tropomyosin (**GO:0005862**), striated-muscle thin filament (**GO:0005865**), actin cytoskeleton (**GO:0015629**).

Metabolic / immune / epigenetic: No primary metabolic defect, no autoimmune/inflammatory mechanism, and no epigenetic mechanism characteristic of TPM2 disease (distinguishing it from acquired sporadic late-onset nemaline myopathy, which has an immune/MGUS association).

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7. Anatomical Structures Affected

Organ level: Primary — **skeletal muscle** (UBERON:0001134). Preferentially **axial, facial, limb-girdle, and respiratory (diaphragm/intercostal) muscles**. Secondary — skeleton (scoliosis, joint contractures), respiratory system (ventilatory insufficiency from muscle weakness), and feeding/bulbar apparatus. **No primary cardiac or CNS involvement.**

Tissue/cell level: Striated skeletal muscle tissue; **slow (type I) skeletal muscle fibers** (CL:0002211) are preferentially affected — TPM2 is *"mainly expressed in slow, type 1 muscle fibers."* Pathology includes type-1 fiber predominance and atrophy.

Subcellular level: Sarcomere / thin filament (GO:0030017, GO:0005865, GO:0005862). Nemaline rods derive from Z-disk/α-actinin material; cap-like lesions are subsarcolemmal thin-filament accumulations.

Localization / lateralization: Generally **bilateral and symmetric**. Distal arthrogryposis is characteristically bilateral/symmetric (clenched fists, overlapping fingers, talipes equinovarus).

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8. Temporal Development

Onset: Typically **congenital or neonatal** (hypotonia, feeding/respiratory difficulty, contractures); some milder cases present in childhood. Prenatal clues include **decreased fetal movements (HP:0001558)** and **polyhydramnios (HP:0001561)**.

Progression: Slow or non-progressive / relatively stable in most cases. Distal-arthrogryposis contractures are non-progressive. The cap-myopathy subset can progress to respiratory failure in infancy (PMID: 25079567 context). Disease is **chronic and lifelong**.

Patterns: No spontaneous remission; course is static-to-slowly-progressive. **Critical period:** The developmental/myogenesis window is mechanistically critical (disrupted myofibrillogenesis), suggesting early muscle development is a key vulnerability period.

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9. Inheritance and Population

Epidemiology: Congenital myopathies as a group: **birth prevalence 14.9/100,000, cumulative incidence by age 35 of 12.3/100,000, point prevalence 3.9/100,000** (western Sweden) (PMID: 41159764). TPM2 is a **minority cause**. In distal arthrogryposis, mutations were *"distributed nearly equally across TNNI2, TNNT3, TPM2, and MYH3"* (PMID: 23401156) — TPM2 is one of four roughly equal DA genes.

Inheritance: Predominantly **autosomal dominant** (de novo or inherited); rare **autosomal recessive** (Escobar-variant). **Penetrance** is high for the dominant form; **expressivity is variable** (same gene → NM/cap/core-rod/CFTD/DA). No genetic anticipation (not a repeat-expansion disorder). Germline mosaicism is plausible but not well documented. **Carrier frequency** of pathogenic LoF alleles is low; heterozygous LoF carriers are unaffected (recessive form). **Consanguinity** is relevant only to the recessive form.

Population demographics: No strong ethnic predilection; cases reported worldwide (European, Chinese, Middle Eastern cohorts). No established sex bias. Founder effects are not described for TPM2; the recessive Escobar variant arose in a consanguineous family.

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10. Diagnostics

Diagnostic pillars:

1. **Muscle biopsy histopathology** — the defining investigation. *"The condition is defined by the histopathological finding of nemaline bodies (rods) on muscle biopsy"* (PMID: 27659899). Findings: nemaline rods (Gomori trichrome red, electron-dense Z-disk-derived bodies; HP:0003798), cap-like subsarcolemmal structures, central cores/mini-cores, congenital fiber-type disproportion (type-1 fibers 12–25% smaller than type-2) with type-1 predominance. **α-actinin immunohistochemistry** highlights rods/caps.

2. **Electrophysiology / labs** — myopathic EMG (HP:0003458); CK normal or mildly elevated (HP:0008180). In cap myopathy with myasthenic features, repetitive nerve stimulation/ single-fiber EMG may reveal a transmission defect.
3. **Molecular genetics** — targeted **NGS congenital-myopathy / nemaline gene panels** (TPM2, TPM3, ACTA1, NEB, TNNT1, KBTBD13, RYR1, etc.) or **whole-exome / whole-genome sequencing**; single-gene TPM2 testing when phenotype is characteristic (trismus-pseudocamptodactyly, hypercontractile DA). **Linkage** can map to TPM2 (PMID: 30285720). **Intronic/splice variants** (c.564-2A>C) may require **WGS/RNA analysis** beyond exon panels (PMID: 33558124).

Imaging: Muscle MRI can show selective muscle involvement patterns (supportive, not specific).

Differential diagnosis: Other congenital myopathies (TPM3, ACTA1, NEB, KBTBD13/NEM6, RYR1), congenital muscular dystrophies (LAMA2-RD), other distal-arthrogryposis genes (TNNI2, TNNT3, MYH3, PIEZO2), and acquired sporadic late-onset nemaline myopathy (SLONM — distinguished by late onset, MGUS/HIV association, treatable).

Screening: No newborn/population screening. **Cascade genetic testing** of at-risk relatives and **prenatal/preimplantation testing** are feasible once the familial variant is known.

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11. Outcome / Prognosis

Survival / mortality: Generally **normal-to-near-normal life expectancy**, especially for distal-arthrogryposis and milder myopathy forms. **Respiratory muscle weakness is the principal threat**; the cap-myopathy subset can develop infantile respiratory failure.

Morbidity / function: In a cross-sectional nemaline-myopathy study (pooled NM genes, not TPM2-only): *"Fifty-eight percent of individuals needed mechanical support, with 26% requiring wheelchair, tracheostomy, and feeding tube"* (PMID: 33397769); course was *"relatively stable."* TPM2 disease is at the **milder** end: *"Patients with TPM2 mutations tended to present with milder symptoms than those with TPM3 mutations"* (PMID: 24692096).

Complications: Respiratory insufficiency, scoliosis, contractures/orthopedic deformity, feeding difficulty/failure to thrive, and (rarely) secondary neuromuscular-transmission defects.

Prognostic factors: Histological subtype (cap → respiratory risk), degree of respiratory/bulbar involvement, and (experimentally) variant severity in myogenesis assays, which correlated with clinical severity in DA patients (PMID: 35579956).

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12. Treatment

No approved disease-modifying therapy. Management is **supportive and multidisciplinary** (NCIT concepts: supportive care, physical therapy, respiratory therapy, orthopedic surgery, nutritional support):

- **Respiratory care** — surveillance of pulmonary function, non-invasive/invasive ventilation as needed; the main determinant of survival.
- **Orthopedic/rehabilitative** — physiotherapy and occupational therapy for contractures and mobility; surgical correction of scoliosis, talipes, and fixed contractures; serial casting/bracing in distal arthrogryposis.
- **Nutritional/GI** — management of feeding difficulty, dysphagia, reflux; gastrostomy when required.
- **Pharmacotherapy** — no specific drug therapy. A **treatable exception**: cap myopathy with a secondary neuromuscular-transmission defect responds to **anticholinesterase therapy** (e.g., pyridostigmine) — *"We describe a case of cap myopathy with myasthenic features due to a mutation in the TPM2 gene that responded to anticholinesterase therapy"* (PMID: 25127990).
- **Pharmacogenomics / gene / cell / RNA therapy** — none established; allele-selective silencing is conceptually attractive given LoF tolerance (see follow-up).

Treatment strategy: Genotype/phenotype-guided surveillance (respiratory emphasis for cap myopathy; orthopedic emphasis for DA). No combination or targeted regimens exist.

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13. Prevention

Primary prevention: Not possible for a constitutional genetic disorder beyond reproductive options. **Genetic counseling** (autosomal dominant 50% transmission risk; recessive 25% risk in consanguineous families) and **preimplantation/prenatal genetic testing** once the familial variant is known.

Secondary prevention: Early multidisciplinary surveillance (respiratory, orthopedic, nutritional) to detect and manage complications early.

Tertiary prevention: Prevention of complications — respiratory support to avert failure, scoliosis/contracture correction, aspiration/nutrition management.

Screening/counseling: Cascade testing of relatives; genetic counseling per NSGC/ACMG. No immunization or public-health/environmental interventions apply.

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14. Other Species / Natural Disease

Taxonomy / orthologs: Human *TPM2* is evolutionarily conserved. Orthologs — mouse *Tpm2* (NCBI Gene **22004**), rat *Tpm2* (**500450**), zebrafish *tpma* (**30324**), *Drosophila* tropomyosin Tm1/Tm2; conserved coiled-coil actin-binding thin-filament protein (HomoloGene 134045).

Natural disease / veterinary relevance: No naturally occurring TPM2 disease is catalogued in OMIA. There is no documented companion-animal or wildlife spontaneous TPM2 myopathy; all animal disease is **engineered/induced**.

Comparative biology: The thin-filament contractile mechanism is deeply conserved from invertebrates to mammals, which is why fly/fish/mouse models recapitulate human TPM2 pathology. **Zoonotic/cross-species transmission:** not applicable (non-infectious genetic disease).

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15. Model Organisms

Model types and genetic models: Pathogenicity of dominant TPM2 variants has been validated across three systems through **transient overexpression/transgenesis** (PMID: 35579956):

Model	System	Finding
Drosophila embryo	Invertebrate, in vivo	4 variants significantly affected muscle development & function
Mouse myotube (C2C12-type)	Mammalian, in vitro	Variants disrupted myotube morphogenesis
Zebrafish	Vertebrate, in vivo	Impaired muscle development; **severity correlated with clinical severity** (V129A, E139K, A155T)

An earlier **zebrafish p.K7del model** showed the mutant protein *"fails to localize properly within the thin filament compartment and its expression alters sarcomere length"* (PMID: 23413262). **In vitro reconstitution / motility assays** with purified human β-tropomyosin quantify altered Ca^{2+} sensitivity (R91G, PMID: 29097206; Q93H/E97K, PMID: 33919826; ΔK7 and motif mutations, PMID: 23886664).

Model characteristics: These models recapitulate developmental muscle defects, thin-filament mislocalization, altered sarcomere length, and calcium-handling abnormalities, and predict clinical severity. **Limitations:** no established mammalian **knock-in** model of a recurrent allele carried to adulthood; most assays use transient overexpression rather than physiological endogenous expression, and do not fully model the mature-muscle, lifelong clinical course.

Resources: MGI (mouse Tpm2), RGD (rat), ZFIN (zebrafish tpma), FlyBase (Tm1/Tm2).

Mechanistic Model / Interpretation (synthesis)

Genotype → molecular phenotype → clinical phenotype (illustrative)

Variant	Molecular effect	Predominant clinical phenotype
ΔK7 (p.K7del)	↑ Ca^{2+} sensitivity; disrupts N-terminal polymerization	Core-rod myopathy; trismus-pseudocamptodactyly (DA7)
R91G	High Ca^{2+} sensitivity	Distal arthrogryposis
E139K / ΔE139	↑ Ca^{2+} sensitivity; disrupts myogenesis	Distal arthrogryposis / congenital myopathy
R167H	No change in Ca^{2+} sensitivity	Non-hypercontractile myopathy
c.564-2A>C (homozygous)	Loss of exon-6b isoform (recessive LoF)	Escobar-variant MPS + congenital myopathy

The data converge on a **two-branch gain-of-function model**: dominant missense/in-frame alleles perturb the actin–tropomyosin interface, producing either **hypercontractility** (→ contractures/distal arthrogryposis) or **disrupted myofibrillogenesis** (→ structural lesions and weakness), often both. This is independently corroborated by population genetics (missense-constrained, LoF-tolerant) and by cross-species models whose severity predicts clinical severity. A minor, mechanistically opposite recessive LoF branch produces the Escobar-variant phenotype.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
24692096	*Mutation update & genotype–phenotype, TPM2/TPM3*	Human, 53-family review	Spectrum; hypercontractility–contracture link; milder than TPM3
23886664	*Mutations in repeating motifs cause gain of function*	In vitro biochemistry	↑ Ca^{2+} sensitivity; actin–tropomyosin interface
35579956	*TPM2-directed myogenesis pathomechanism*	Fly/mouse/fish models	Disrupted myogenesis; severity correlation
33558124	*Homozygous intronic TPM2 → Escobar + myopathy*	Human, consanguineous family	Recessive LoF; splice variant
32092148	*Recurrent TPM2 variant in MPS-related disorder*	Human, 3 cases	De novo dominant MPS
23401156	*Mutations in DA1 and DA2B*	Human, 153 kindreds	TPM2 one of four DA genes
41159764	*Incidence/prevalence of congenital myopathy, Sweden*	Population-based	Epidemiology
25127990	*Congenital myopathies with secondary NMJ defects*	Human case + review	Treatable NMJ defect
27659899	*Nemaline myopathies: state of the art*	Review	Biopsy definition
33397769	*Cross-sectional study of nemaline myopathy*	Human cohort	Morbidity/support needs
23413262	*Novel K7del expands TPM2 spectrum*	Human + zebrafish	Core-rod; DA7; polymerization defect
29097206	*Reason for high Ca^{2+} sensitivity (R91G)*	In vitro	R91G biochemistry
33919826	*Mutations Q93H and E97K*	In vitro	Ca^{2+} regulation

PMID	Title (abbrev.)	Evidence type	Supports
30285720	*Novel TPM2/PIEZO2 in DA*	Human, linkage + WES	Linkage mapping; no position–phenotype rule
25079567	*Cap-like structures and nemaline rods*	Human case + review	Cap myopathy natural history (respiratory risk)

Source-type triangulation: human clinical/genetic (majority), in vitro biochemistry (mechanism), multi-species model organisms (developmental mechanism), and gnomAD population constraint. The gain-of-function conclusion is supported by all four independent evidence classes, giving it high confidence.

Limitations and Knowledge Gaps

1. **No TPM2-specific natural-history cohort.** Prognostic data are borrowed from mixed nemaline-myopathy cohorts; TPM2-only survival curves and QoL metrics (EQ-5D/SF-36/PROMIS) are unavailable.
2. **High VUS burden** (~285/480 ClinVar records), limiting clinical interpretation; functional classification is not scaled to the full variant set.
3. **Mechanistic branch weighting unknown** — the relative contribution of hypercontractility vs disrupted myogenesis per patient is not quantified.
4. **Modifier genes and epigenetics uncharacterized** for TPM2 disease.
5. **No approved disease-modifying therapy**; the anticholinesterase response is based on isolated cases.
6. **Imperfect genotype–phenotype prediction** — one study found no relation between variant type/location and clinical features.
7. **Underdiagnosis risk** from deep-intronic/splice variants missed by exon-only panels.

Proposed Follow-up Experiments / Actions

1. **Build a TPM2-specific natural-history registry** (onset, motor/respiratory trajectory, scoliosis, survival) stratified by histological subtype and hypercontractile vs non-hypercontractile allele class.
2. **High-throughput VUS reclassification** using standardized Ca^{2+}-sensitivity motility and zebrafish/myotube myogenesis assays; feed results to ClinVar/ClinGen.
3. **Systematic screen for secondary NMJ defects** (RNS, single-fiber EMG) across the cohort to identify anticholinesterase-responsive patients — low-cost, potentially practice-changing.
4. **Preclinical testing of calcium-desensitizing agents** in hypercontractile TPM2 models, targeting the defined druggable node of excess myofilament Ca^{2+} sensitivity.
5. **Allele-selective RNA therapeutics (ASO/siRNA)** to silence the dominant mutant allele, leveraging gnomAD LoF tolerance (haploinsufficiency is not disease-causing).
6. **Routine WGS + RNA-seq** for biopsy-positive, panel-negative congenital myopathy to capture splice variants like c.564-2A>C.
7. **Generate a knock-in mouse** carrying a recurrent hypercontractile allele (R91G or E139K) for in-vivo therapeutic and natural-history studies.

Report generated from a 5-iteration autonomous investigation: 13 confirmed findings, 30 papers reviewed. Evidence classes: human clinical/genetic, in vitro biochemistry, multi-species model organisms, and population-genetic constraint.
